

Amrut Savadatti

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Summary

Full-Stack Software Engineer with **4+ years of experience** building & operating production web applications and APIs upholding clean code practices

EXPERIENCE

Web Application Developer

Aug 2021 - May 2024

Media.net

- Designed and developed a push-based Keyword Launcher API in Java, Spring Boot, and MongoDB to automate keyword compliance checks, **reducing latency by 35%** and enabling real-time analytics.
- Implemented centralized JWT authentication using Spring Security to secure 30+ APIs, improving security consistency and streamlining onboarding for new microservices.
- Maintained and developed features for core Arbitrage Plugin Backend serving **400+ publisher sites**, enabling dynamic ad configuration, A/B testing, and click tracking to optimize monetization across ad networks using PHP and MySQL.
- Architected a centralized API Proxy Service with secret-key management, response caching, and Dockerized deployments, **reducing redundant third-party API calls by 30%**.
- Owned end-to-end containerization and deployment of internal tools: Dockerized applications, integrated CI/CD pipelines with SRE teams, and deployed to AWS EKS with load testing via K6 and monitoring through Grafana, ELK, and Sentry.

Full-Stack Software Engineer - Beecology

Sep 2025 - Present

WPI + Umass Dartmouth

- Took ownership of a legacy Angular application with a broken Virtualearth Elevation API dependency, migrated the frontend to React.js and Tailwind CSS, and extended the Flask backend with new RESTful endpoints and PostgreSQL integration.
- Delivered performant UI components for video playback, media uploads, ML-driven species recognition, and interactive map-based data collection used by researchers across WPI and UMass Dartmouth.
- Dockerized and deployed the full-stack application serving **5,000+ field observations** per month documenting elevation, geolocation, and seasonal activity patterns for bumblebees and butterflies to support ecological behavioral research.

SKILLS

- Frontend & Mobile::** React, React Native, JavaScript, TypeScript, Next.js, Tailwind CSS
- Backend:** Node.js, Flask, FastAPI, Spring Boot, Spring Security, Java, Python, PHP
- Databases & Cloud:** MySQL, Redis, AWS, MongoDB, Postgres
- DevOps & Tooling:** Docker, Git, ELK, Grafana, LAMP, Kubernetes
- AI skills:** RAG, GAN, Transformers, Claude Skills

PROJECTS

AI Portfolio Chatbot | <https://github.com/amrutsavadatti/AI-assistant>

Jul 2025 - Aug 2025

- Built a Retrieval-Augmented Generation (RAG) AI assistant using LangChain, ChromaDB, FastAPI, and OpenAI APIs to answer natural language queries about professional background, projects, and technical skills.
- Designed and iterated on the portfolio frontend experience, integrating 2FA-verified email capture and Calendly scheduling to streamline recruiter engagement.
- Containerized and deployed the full-stack application using Docker Compose on AWS EC2, with per-user rate limiting to prevent API abuse and ensure reliable performance.

Custom AI Tool Server

Dec 2025 - Dec 2025

Worcester Polytechnic Institute

- Built a custom MCP server in Python using FastAPI for transport, Pydantic for schema validation, and the mcp library's `@server.tool` decorator to define extensible tool endpoints.
- Implemented three integrated productivity tools weather lookup via third party APIs, a statistical calculator using NumPy and SciPy, and a Google Calendar integration with OAuth 2.0 for automated event creation.

EDUCATION

Worcester Polytechnic Institute

Aug 2024 - Aug 2026

Masters of Science, Computer Science

- Coursework:** Operating Systems, Artificial Intelligence, System Design, MCP server, Generative AI

University of Mumbai

Aug 2017 - Jun 2021

Bachelor of Engineering, Computer Engineering

- Coursework:** ML, Database Management Systems, Object oriented programming, Algorithms, Data Structures